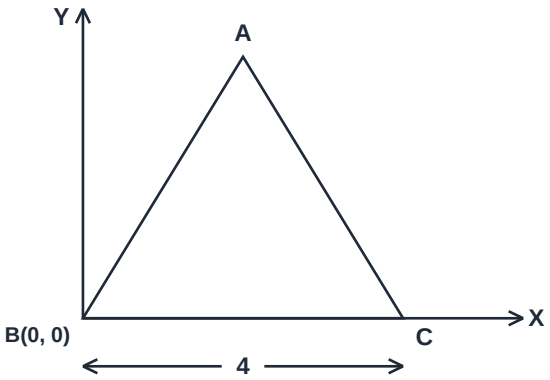
COORDINATES

PYQ • 2021 - 2026

1. 2026 **3 MARKS**

In the figure, triangle ABC is equilateral.



(i) What are the coordinates of C? **(1)**

(ii) Calculate the height from A to BC. **(1)**

(iii) What are the coordinates of A? **(1)**

2. 2026 **4 MARKS**

(A) A circle is drawn with the line joining the points (3, 5) and (5, 9) as diameter.

(i) What are the coordinates of the centre of the circle? **(1)**

(ii) What is its radius? **(1)**

(iii) Write the equation of the circle. **(2)**

OR

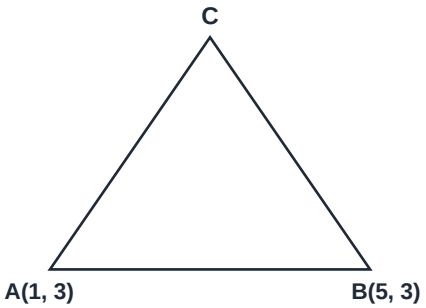
(B) (i) Find the slope of the line passing through the points (3, 5) and (9, 7). **(1)**

(ii) Write the equation of the line. **(2)**

(iii) Write the coordinates of any other point on this line. **(1)**

3. 2025 **4 MARKS**

In the figure, triangle ABC is an equilateral triangle. A is (1, 3) and B is (5, 3).



(a) Find the length of AB.

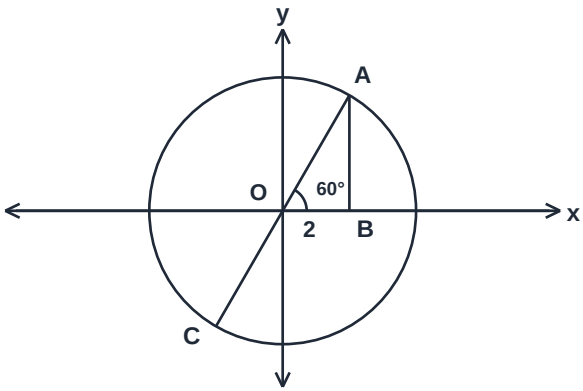
(b) Write the coordinates of the midpoint of AB.

(c) Find the coordinates of the vertex C.

4. 2025 **4 MARKS**

In the figure, a circle is drawn with the origin as the centre. A is a point on the circle.

Angle AOB = 60°; OB = 2 units.



(a) Write the coordinates of A.

(b) What is the radius of the circle?

(c) Write the equation of the circle.

(d) Write the coordinates of the point C on the circle.

5. 2024 **4 MARKS**

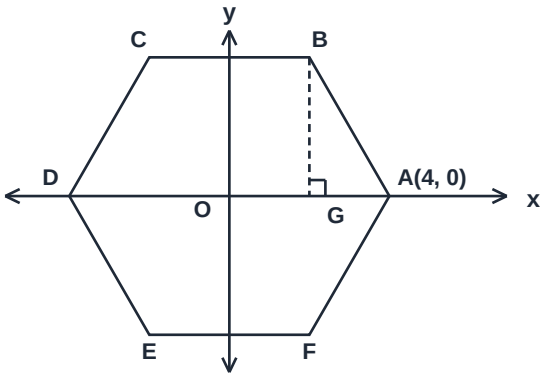
A(2, 3), B(8, 5) and C(4, 7) are the coordinates of the vertices of triangle ABC. P is the midpoint of AB and Q is the midpoint of BC.

(a) Find the coordinates of P and Q.

(b) Find the distance between P and Q.

6. 2024 **4 MARKS**

ABCDEF is a regular hexagon with the origin as centre. The coordinates of the point A is (4, 0).



(a) What are the coordinates of the point D?

(b) Find the length of BG.

(c) Write the coordinates of the points B and E.

7. 2024 **5 MARKS**

The equation of a line is $2x - y - 2 = 0$.

(a) Check whether the point (3, 4) is on this line.

(b) Find the coordinates of the points where this line cuts the x and y axes.

8. 2023 **3 MARKS**

A(3, 9), C(8, 12) are the coordinates of two opposite vertices of a rectangle whose sides are parallel to the coordinate axes.

(a) Find the coordinates of other two vertices of the rectangle.

(b) Find the lengths of the sides of the rectangle.

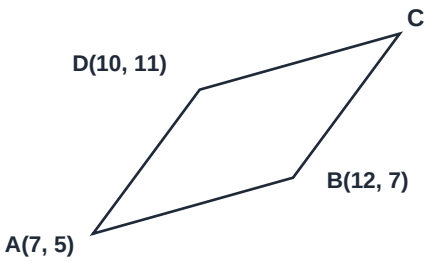
9. 2023 **3 MARKS**

(4, 5) and (8, 11) are coordinates of two points on a line.

(a) Find the slope of the line.

(b) Find the equation of the line.

10. 2023 **4 MARKS**



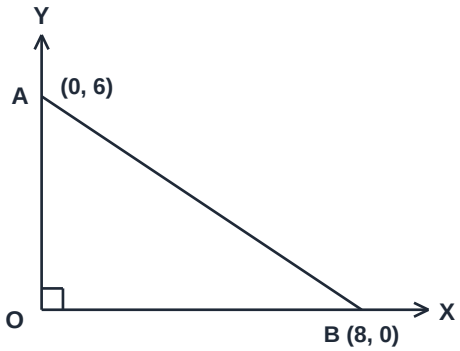
In the figure, co-ordinates of 3 vertices of the parallelogram ABCD are given.

(a) Write the co-ordinates of C.

(b) Calculate the length of the diagonal AC.

(c) Find the co-ordinates of the point of intersection of the diagonals.

11. 2023 **4 MARKS**



In the figure, (0, 6) and (8, 0) are coordinates of the points A and B. A circle of diameter AB is to be drawn.

(a) Find the coordinates of the centre of the circle.

(b) Find the radius of the circle.

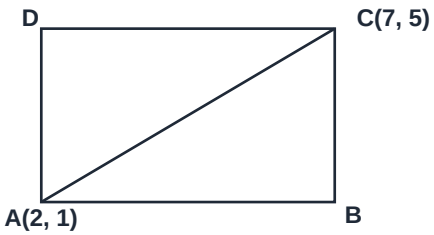
(c) What is the equation of the circle?

12. 2022 **1 MARK**

The equation of a line is 2x + y = 5. If the x co-ordinate of a point on this line is 2, what is the y co-ordinate of this point?

(0; 1; –1; 2)

13. 2022 **6 MARKS**



The rectangle has sides parallel to the axes. The co-ordinates of one pair of opposite vertices are (2, 1) and (7, 5).

(a) Find the co-ordinates of the other two opposite vertices. **(2)**

(b) Find the length and breadth of the rectangle. **(2)**

(c) Find the length of the diagonal AC. **(2)**

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14. 2022 **8 MARKS**

(a) Draw the coordinate axes and mark the points (2, 1) and (4, 3). **(3)**

(b) Find the slope of the line joining these points. **(2)**

(c) The centre of a circle is (3, 2) and the coordinates of one end of its diameter is (1, 2). Find the coordinates of the other end of the diameter. **(3)**

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15. 2021 **1 MARK**

What is the slope of the line passing through the points (2, 5) and (3, 7)?

(2; 3; 4; 5)

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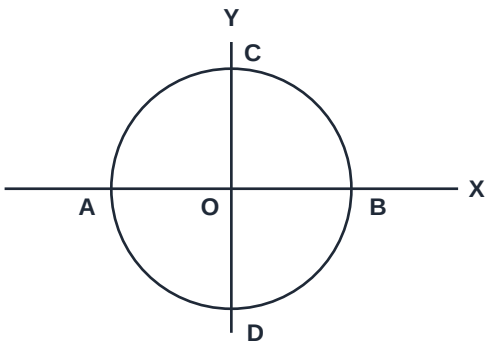
16. 2021 **2 MARKS**



In the figure, AB and CD are diameters of the circle. Coordinates of B are (3, 0). Write the coordinates of O and C.

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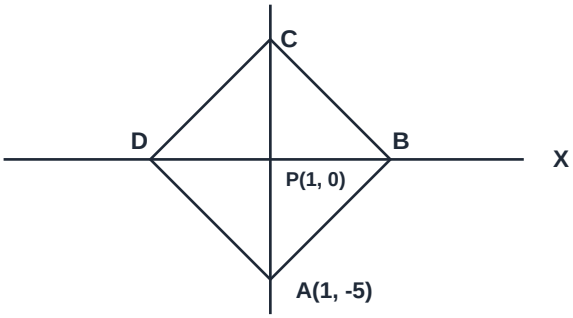
17. 2021 **3 MARKS**

ABCD is a square, coordinates of A are (1, –5). Diagonals of the square intersect at P(1, 0). Write the coordinates of B, C and D.



18. 2021 **4 MARKS**

Draw the x and y axes. Mark the point (2, 3). Draw a circle with origin as centre and passing through the point (2, 3).

19. 2021 **5 MARKS**

The sides of a rectangle are parallel to the axes. One pair of its opposite vertices are A(2, 4) and C(6, 12).

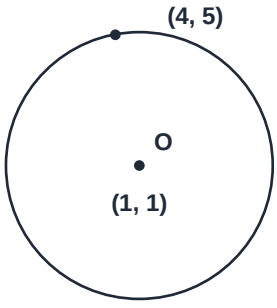
(a) Write the coordinates of the other two vertices. **(2)**

(b) Write the coordinates of the mid-point of AC. **(2)**

(c) x coordinate of a point on AC is 'a'. What is its y coordinate? **(1)**

20. 2021 **5 MARKS**

A circle is drawn with (1, 1) as centre. (4, 5) is a point on the circle.



(a) Find the radius of the circle. **(1)**

(b) Write the equation of the circle. **(2)**

(c) The x coordinate of a point on the circle is 6. What is the y coordinate of that point? **(2)**